Validation and AI use

AI-GENERATED DRAFT | October 19, 2026 | Student review pending

Decision, version and boundary

Proposed watch/defer for a committee with no SMR position. Analysis boundary: September 1, 2026 close, later than the pictured Edition A lab time. Research and draft construction occurred October 19. No September authorship is claimed. The working record matches the generated draft PDF when rebuilt.

Core environment: Python 3.10+ and standard library. PDF rebuilding uses optional packages in requirements-docs.txt. Input and source SHA-256 hashes are embedded in output/commercialization-output.json. The package file inventory provides file hashes; no Git freeze or LMS receipt is invented.

Data and convention ledger

Historical observations, units, dates and filing locators are in docs/verified-starting-point-draft.md and data/research-inputs.json. These are reported June balances, June awards and July cover shares, not September balances. The model's assumed $40m lag-period cash use reduces cash and opening equity. The unrestricted liquidity sum adds each selected category once; restricted balances are outside it and are not subtracted a second time.

Current valuation uses consolidated economic interests with a gross award allowance. No book-NCI subtraction is made again. Gross options with no proceeds are an allowance, not a treasury-stock-method calculation. Equity funding affects liquidity; its fixed-price dilution diagnostic is separate from FCFF valuation.

Operating commitments and leases are assumed inside forecast costs. This requires a contract-by-contract mapping before a final value. Additional claims default to zero as a scenario condition, not a legal conclusion. The engine uses economic PMA expense timing, not GAAP revenue recognition or a tax opinion.

Source check and assistant correction

The assistant opened the PMA exhibit, Sections 1(c), 1(d), 1(g), and found the annual escalation floor and redacted cap. Earlier flat-fee outputs were preserved and superseded. The revised fee scale is inferred from the 2025 cash installment and installment allocation. Future fees grow at the 5% floor; higher CPI is not forecast. Lead-year timing simplifies contractual slots, invoices and contingencies.

An unsupported low-growth perpetuity is disabled. Default operating exit value is six times positive normalized next-year EBITDA after full milestone fees. The multiple is an analyst assumption, not a peer benchmark. Zero/4x/6x/8x/10x alternatives are computed. Negative EBITDA receives zero operating sale value, not a liquidation appraisal. Quantitative peer validation remains unresolved.

Validation register

Claim	Check	Actual disposition	Evidence
Cash is not counted twice	No-operations known answer	PASS	tests/test_commercialization.py
Cash follows financial flows	Independent cumulative cash budget	PASS	same test file
Statements reconcile	Assets equal claims; equity rolls from income/funding	PASS	same test file

Claim	Check	Actual disposition	Evidence
Sunk tranche is not paid twice	Single-cohort hand formula with escalation	PASS	same test file
Contract escalation matters	Compare escalated vs flat fees	PASS	same test file
Funding is not double charged	Financing-price change affects diagnostic shares only	PASS	same test file
Margin stress worsens economics	Unfloored equity falls; funding does not fall	PASS	same test file
Discount dates are explicit	Recompute actual-day discount factors	PASS	same test file
Invalid assumptions fail	Reject bad rates, deliveries and nonfinite inputs	PASS	same test file
Unsupported terminal path fails	Reject growth below fee escalation if enabled	PASS	same test file
UI displays executed values	Compare CLI summary with frozen output	PASS	tests/test_application.py
UI detects stale evidence	Alter copied input and require error	PASS	same test file
Invalid year is actionable	Require nonzero exit and valid range	PASS	same test file
Old downside invariants	Cash budget, bridge, funding, claims, broken balance	PASS: five checks	tests/test_smr_proforma.py

Run python3 verify_project.py. Current suite: 20 tests passed. Tests demonstrate these specific invariants, not the accuracy of commercial forecasts.

Failure retained and diagnosed

After fee escalation was corrected, one margin test initially failed because it required strictly lower floored share value even when both outcomes were zero. Diagnosis: limited-liability flooring masks a worse underlying equity result. The corrected test requires lower unfloored equity, nonincreasing floored value, and no reduction in funding. No finance input was tuned to force a passing test.

Changed-input results - AI-generated, not the locked exercise

The three five-percentage-point module-margin stresses and the full price/discount grid are preserved in output/commercialization-output.json. Stronger execution moves from $6.77 to $3.34/share including the assumed operating sale. The watch/defer proposal remains because contract pricing and the assumed exit are unverified, not because the automated test establishes a final price target.

Reproducibility and access

evidence/clean-run.md records an actual assistant rehearsal in a fresh temporary directory with site packages disabled. Its command outputs and JSON comparisons are preserved. This is not a claimed outside-person cold run. A real reviewer must record their own first failure/resolution if the assignment requires it. Repository publication state is recorded in evidence/repository-status.json. Instructional-team and video access remain separate checks; they are not inferred from an assistant's successful git connection.

Existing NuScale Lab 11 changed-input evidence

The exercise is not classified as absent. Existing records include a supplied prediction, input/model snapshot, changed cost-growth input, before/after output, and reconciliation. They are preserved in evidence/lab11-smr/ and mapped in docs/lab11-evidence-map.md. Cost growth changes from 0% to 2%; 2030 FCFE moves from -174.639m to -196.199m. The recorded explanation attributes the 21.560m decline to 20.028m extra costs and 1.532m lost investment income. Watch/defer remains.

The saved narrative does not independently establish the prediction-lock timestamp or AI boundary during the original run. This is a documentary check on existing work, not an automatic requirement to repeat it. Joby comparison evidence is not substituted for NuScale ownership. No timestamp or attestation is invented.

Named AI-use register

Tool: Codex; exact model version was not exposed in the task surface. Interaction date: October 19, 2026. Material uses: workspace audit, source navigation, code, scenario assumptions, tests, generated outputs, memo/research/validation drafts, PDF layout, application interface and rehearsal scripts.

Assistant disposition: modified the flat-fee proposal after checking the contract; withdrew unsupported perpetual terminal credit; qualified every module-price and delivery assumption. Primary-source checks and deterministic arithmetic are assistant work. Student accept/modify/reject decisions and independent checks are still pending and must be recorded truthfully.

Inherited Week 5 work is preserved. Its original AI history must be supplied by the student. This record does not infer authorship from a test pass, file date, lab grade or code style. The user's account of grammar assistance is retained as their account, not independently verified history.

Limitations, monitoring and kill rules

See docs/monitoring-policy.md. The full event/share audit, net contract economics, invoice timing, investment liquidity, authorized share capacity, contingent/tax claims and residual valuation remain open. Assumed equity issuance is not committed capacity, and end-year cash checks cannot establish intra-year solvency. No default, legal liability or certain loss is inferred from a model flag.

Source references

PMA: https://www.sec.gov/Archives/edgar/data/1822966/000182296625000144/a101-partnershipmileston.htm June 10-Q: https://www.sec.gov/Archives/edgar/data/1822966/000182296626000085/smr-20260630.htm Source ledger and boundary review: docs/verified-starting-point-draft.md; docs/source-review.md.